TR-Hash 200M: Multi-Hash Token Routing Across 162B Token Exposures and Full-Parameter SFT

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Abstract

We report the released training lineage of TR-Hash 200M, a 201.2M-parameter decoder-only language model whose feed-forward blocks combine an always-on shared SwiGLU path with four deterministic token-ID-routed residual experts, two active per token. Routing is compiled once per layer by multi-hash rendezvous voting; it has no learned gate and adds no runtime routing model. The lineage comprises (i) a completed 129.996B-token replay-scheduled base pretraining run, (ii) a fresh-optimizer full-parameter refinement stopped at step 8,156 after 32.069B additional unique-token exposures, and (iii) three epochs of full-parameter supervised fine-tuning (SFT) over 238.9M supervised tokens. On the full 1,838-example PIQA validation split, zero-shot causal continuation scoring without a chat template moves from 65.45/65.61 accuracy/length-normalized accuracy for the base checkpoint to 68.66/68.39 after refinement and 68.82/69.31 for the promoted SFT epoch-2 checkpoint. Matched held-out SFT loss decreases from 1.7628 before SFT to 1.2209 after epoch 3. These are single-lineage results without a parameter-matched dense or learned-router control, multi-seed replication, or contamination audit; they establish a reproducible released artifact, not architectural superiority.

1 Introduction

Learned mixture-of-experts (MoE) routers condition parameter selection on the current hidden state, but quality, capacity, dispatch, router optimization, and balancing objectives then change together [7, 4, 5]. Fixed token-ID routing is a deliberately narrower intervention: attention continues to build contextual hidden states, while lexical identity selects a small residual feed-forward subspace. The selection can be persisted, audited, and reproduced independently of activations or training progress.

This paper updates the earlier architecture-and-plan report for TR-Hash 200M. That report intentionally contained no training result because the base run was unfinished. The present version records the realized three-stage lineage, the released checkpoint selected after full-parameter SFT, and the limits of the evidence. It does not fold the earlier experimental LoRA work into the release: all weights in the promoted assistant were updated by full-parameter SFT.

Table 1: Released architecture. Stored parameters include all four routed experts; two are active for each token.

Component	Value
Trainable parameters	201,194,368
Layers / hidden size	16 / 896
GQA query / KV heads	14 / 2
Vocabulary / context	32,000 / 2,048
Shared SwiGLU width	3,072
Stored experts / width	4 / 256
Active experts	top-2
Routing channels	2 hashes

The contributions are:

1. an exact description of the released 201.2M multi-hash shared-plus-residual architecture;
2. an auditable accounting of 129.996B base exposures, 32.069B refinement exposures, and 238.9M supervised SFT tokens;
3. checkpoint-level PIQA results under one fixed zero-shot continuation protocol; and
4. explicit separation of training minibatch loss, matched SFT validation loss, downstream accuracy, and qualitative chat behavior.

2 Architecture

2.1 Contextual backbone

The model is a pre-norm causal decoder with 16 Transformer layers, hidden size 896, RMSNorm, rotary position embeddings, query/key normalization, and tied input/output embeddings. Grouped-query attention (GQA) uses 14 query heads and two key/value heads [1, 8]. The vocabulary has 32,000 entries and the configured maximum sequence length is 2,048 tokens.

2.2 Shared and routed computation

Let x_l be a contextual hidden state at layer l, t its token ID, S_l the shared SwiGLU branch, and $E_{l,e}$ one of four residual experts. A persisted table $R_l \in \{0, 1, 2, 3\}^{2\times 32{,}000}$ supplies two distinct experts. The realized feed-forward output is

$$F_l(x_l, t) = S_l(x_l) + 2\sum_{j=1}^{2} 0.5\, E_{l,R_{l,j}(t)}(x_l). \tag{1}$$

Thus the routed contribution is the sum of the two selected expert outputs. The shared branch remains active for every token and is substantially wider than each expert. Stored intermediate width is $3072 + 4 \times 256 = 4096$; active intermediate width is $3072 + 2 \times 256 = 3584$. Expert tensors account for 5.47% of model parameters.

2.3 Multi-hash route construction

For every layer, token/expert pairs receive two independent 32-bit rendezvous scores. Scores are summed across hash channels, and the two experts with the largest totals become the token's fixed routes. The result is compiled into the ordinary [top-k, vocab] dispatch table. Consequently, multi-hash construction adds no per-token learned routing network and no additional scoring operation during inference.

The exported tables are training-independent. Across the 16 layers, route tables are pairwise distinct and no token selects the same expert twice. In the first layer, expert occupancy is 15,954 / 16,118 / 15,917 / 16,011 of 64,000 total route slots, each within 0.7% of the uniform target. This is marginal balance over the vocabulary, not a claim that ordered expert pairs or runtime corpus traffic are uniform. Fixed lexical hashes as routing signals relate to Hash Layers [6]; the always-on shared path is also consistent with the motivation behind shared-expert MoE designs [3].

3 Training Lineage

3.1 Stage 1: replay-scheduled base pretraining

The base run selects 70B unique tokens from a larger immutable, pretokenized mixture and uses source-specific replay counts to target 130B exposures. Sources include DCLM, FineWeb-Edu-Dedup, Stack-Edu, FineMath, InfiWebMath, and Cosmopedia v2. The run completes 165,298 optimizer steps and 129,995,636,736 token exposures. Its last logged training minibatch has loss 2.652628 and token-level perplexity 14.19 at learning rate 3×10^{-5}. Because this is a final minibatch rather than a held-out corpus estimate, it is reported as an execution record, not as a generalization metric.

3.2 Stage 2: full-parameter refinement

Refinement starts from base weights only, with a fresh optimizer and learning-rate schedule. It revisits the same 70B unique-token selection without replay or augmentation. The run is stopped at step 8,156 of 17,802 after approximately 32,069,495,945 additional source-token exposures (45.8% of the planned pass). The terminal progress display records training loss 2.3208 and perplexity 10.2. The release therefore preserves an evaluated intermediate checkpoint; no process is expected to complete the remaining refinement steps.

Table 2: Matched held-out SFT evaluation. Perplexity is exp(loss).

Checkpoint	Step	Loss	PPL
SFT source	0	1.762797	5.83
Epoch 1	463	1.260493	3.53
Epoch 2	926	1.226242	3.41
Epoch 3	1,389	**1.220861**	**3.39**

3.3 Stage 3: full-parameter instruction SFT

Instruction post-training starts from the step-8,156 refinement checkpoint. The model is trained for three epochs over the Luciole 16-way instruction mixture, totaling 238.9M supervised tokens. The mixture covers instruction following, conversation, reasoning, code, mathematics, science, safety, retrieval, rewriting, and summarization. The `complexity-chat-v2` template is used for SFT and chat serving. All model parameters are optimized; this stage is not LoRA or QLoRA.

The held-out SFT set contains 281,617 supervised tokens and is evaluated at the source checkpoint and each epoch boundary. Table 2 shows monotonic improvement through epoch 3. Checkpoint promotion is nevertheless based on downstream PIQA, not the smallest SFT loss alone, so epoch 2 is copied to the root release.

4 Evaluation

4.1 PIQA protocol

We evaluate the full 1,838-example PIQA validation split [2]. Each candidate ending is scored as a causal continuation of the prompt. Accuracy selects the higher total log-likelihood; normalized accuracy selects the higher length-normalized log-likelihood. The protocol is zero-shot, uses no chat template, and caps sequences at 2,048 tokens. The same checkpoint weights are evaluated in FP16 through MLX. This continuation protocol avoids confounding the result with chat-template behavior.

Refinement provides the largest observed PIQA change relative to the released base: +3.21 points in accuracy and +2.78 points in normalized accuracy. Full SFT retains that improvement and adds +0.16 / +0.92 points relative to refinement. PIQA does not measure instruction following, safety, code generation, or open-ended chat quality; it is used here as a narrow checkpoint-selection sanity check.

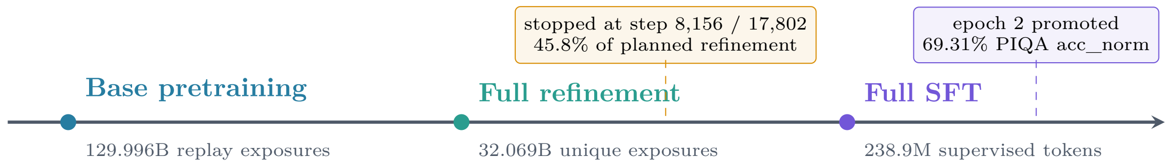

Figure 1: Released lineage. The "160B" repository name is a rounded label for approximately 162.065B source-token exposures; it does not imply that the planned 70B-token refinement completed.

Table 3: PIQA validation results for the released lineage. "Source exposure" excludes the 238.9M supervised SFT tokens. No significance test or multi-seed interval is claimed.

Checkpoint	Source-token exposure	PIQA acc	PIQA acc_norm	Release interpretation
Base final, step 165,298	129.996B	65.45%	65.61%	Completed replay-scheduled base model
Refinement, step 8,156	162.065B	68.66%	68.39%	Interrupted full-parameter intermediate
Full SFT, epoch 2	162.065B + SFT	**68.82%**	**69.31%**	Promoted root assistant checkpoint

4.2 What the losses mean

The three loss signals in this report have different sampling processes. The base and refinement numbers are terminal training displays from autoregressive corpus training. Table 2 is a matched held-out SFT evaluation over supervised labels. Table 3 is a discrete multiple-choice continuation benchmark. A lower SFT loss does not guarantee a higher PIQA score, and none of these numbers alone establishes assistant quality.

5 Implementation and Release Artifacts

The canonical implementation is maintained in the open-source Complexity Framework. The TR-Hash engine exposes a universal PyTorch path, optional Triton/CUDA grouped dispatch, and a CUDA-graph inference path while preserving the same persisted routes. The release includes configuration, tokenizer assets, Safetensors weights, model cards, training logs where available, and an OpenAI-compatible serving stack in TR-Hash-i64.

The promoted root SFT weights have SHA-256

```
2dd5a35624dda4cb267d872f974fd448
c9e085004c163b31772515bfd9573caa.
```

The release is available at https://huggingface.co/AETHORIA-AI/TR-HASH-MoE-200M-160B-SFT; the base and refinement repositories retain their distinct roles. The public chat at https://www.complexity-ai.fr/ai-lab loads the promoted full-SFT model. Experimental LoRA repositories are not part of this lineage.

An interactive 3-D t-SNE artifact probes routed residual contributions on natural PIQA text. It samples two routed contribution vectors per token from layers 1, 4, 8, 12, and 16, excludes the shared branch, L2-normalizes vectors, and applies PCA followed by t-SNE. This is an exploratory inspection tool, not evidence of semantic specialization or quality.

6 Limitations and Responsible Use

This study has one architecture, one training lineage, and no parameter-matched dense or learned-router control. It does not isolate whether the PIQA change comes from additional token exposure, data order, schedule, or the routing architecture. There is no multi-seed replication, statistical significance test, benchmark-contamination audit, memorization analysis, toxicity study, bias audit, or systematic safety evaluation.

The model is small and experimental. It may hallucinate, repeat, refuse benign requests, comply with unsafe requests, emit biased content, fail arithmetic or code, and lose coherence over long contexts. The public chat is a research demonstration, not a reliable agent or decision system. The CC BY-NC 4.0 model license does not replace source-dataset terms; users remain responsible for provenance and compliance.

7 Conclusion

TR-Hash 200M demonstrates that a 201.2M-parameter decoder with fixed, multi-hash token-ID routing can be trained end to end through a long base run, continued with fresh-optimizer full-parameter refinement, and converted into an instruction-tuned assistant by full-parameter SFT. The released epoch-2 checkpoint improves PIQA acc_norm from 65.61% at the base endpoint to 69.31%, while epoch 3 reaches the lowest matched SFT loss. The result is best read as a transparent artifact lineage with precise accounting and known limits, not as proof that deterministic routing outperforms dense or learned alternatives.

Reproducibility. Code: https://github.com/Complexity-ML/complexity-framework. Runtime: https://github.com/Complexity-ML/TR-Hash-i64. Model collection: https://huggingface.co/AETHORIA-AI.

Use of generative AI tools. Generative AI tools assisted with language editing, consistency checks against author-provided configuration and measured artifacts, and PDF packaging. The author reviewed the numerical claims and takes responsibility for the final text.

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